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The Azure AI Foundry Readiness Guide

Is your organisation ready to build AI agents on Azure AI Foundry? This guide covers the five readiness pillars — data estate, Azure environment, use case prioritisation, governance, and team capability — with a self-assessment scorecard included.

Azure AI

Foundry

Copilot Studio

Semantic Kernel

Entra ID

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01 / Introduction

Azure AI Foundry in 2026

Azure AI Foundry is Microsoft's unified platform for building, evaluating, and deploying enterprise AI agents. It consolidates Azure OpenAI Service, Azure AI Search, Prompt Flow LLMOps, the Azure AI Model Catalog (1,600+ models), Semantic Kernel orchestration, and Azure AI Content Safety into a single managed environment with enterprise-grade security, compliance, and cost controls.

For organisations with existing Microsoft 365, Azure Active Directory (Entra ID), and Azure infrastructure investments, Azure AI Foundry is the natural path to production AI agents — the identity, networking, compliance, and data connectors are already in place.

Who this guide is for: CTOs, Azure architects, and AI programme leads at organisations evaluating Azure AI Foundry for production agent deployments. Complete the scorecard in Chapter 7 before any architecture decisions.

The 5 Azure AI Foundry Readiness Pillars

01 Azure Environment

Is your Azure subscription, resource group structure, Entra ID, and networking configured for AI workloads?

02 Data Estate

Is your data in Azure (Blob Storage, Azure SQL, Cosmos DB, Fabric) and accessible to AI agents at query time?

03 Use Case Prioritisation

Have you scored your candidate use cases and identified the highest-ROI pilot to build first?

04 Governance

Are Responsible AI controls, Azure Policy, Entra ID RBAC, Private Endpoints, and audit logging configured?

05 Team Capability

Does your team have the Azure, AI Foundry, and agent engineering skills to build and sustain this in production?

02 / Pillar 1

Azure Environment Readiness

Azure AI Foundry workloads require specific Azure infrastructure configuration before any agent code runs. Environment readiness is consistently the first bottleneck we encounter — most organisations underestimate the baseline configuration work required.

Subscription & Resource Group Structure

Use separate resource groups for dev, staging, and production AI Foundry workloads. Mixing AI workloads with other Azure resources in production increases blast radius for IAM misconfigurations and complicates cost allocation.

Azure AI Foundry Hub & Project

Azure AI Foundry organises work into Hubs (shared infrastructure — compute, storage, key vault, connections) and Projects (agent configurations, deployments, evaluations). The Hub is the organisational unit for governance; Projects are per-use-case.

Entra ID Configuration

Managed Identity is the recommended identity method for Azure AI Foundry workloads — no secrets, no rotation risk, RBAC-governed. Service principals are acceptable but require rotation policy. User-assigned Managed Identities are preferred for cross-resource access patterns.

Model Deployments

Azure OpenAI models (GPT-4o, GPT-4o-mini, GPT-4-turbo) must be deployed via Azure OpenAI Studio before use in AI Foundry. Model availability varies by region — check region-model availability against your data residency requirements before choosing a deployment region.

Private Endpoints

For production deployments, Azure AI Foundry components (Azure OpenAI, Azure AI Search, Storage) should communicate via Private Endpoints — eliminating public internet exposure. Private Endpoint configuration adds 1–2 days of networking work but is non-negotiable for regulated industries.

Quota & Rate Limits

Azure OpenAI has regional TPM (tokens per minute) quotas. Request quota increases 1–2 weeks before production go-live. Insufficient quota causes 429 errors in production that are difficult to diagnose as quota-related.

03 / Pillar 2

Data Estate Assessment

Azure AI Foundry agents reason over data. Where that data lives, how clean it is, and how it is indexed for retrieval determines agent accuracy. Data estate readiness is the pillar where the gap between "we have data" and "our agents can use this data" is widest.

Data Location

Azure AI Search can index data from Azure Blob Storage, Azure SQL, Cosmos DB, SharePoint Online, and website content. Data in on-premise systems or AWS/GCP requires migration or connector setup before agents can access it.

Microsoft Fabric Integration

For organisations with Microsoft Fabric (OneLake), Azure AI Foundry integrates directly — agents can ground against Fabric lakehouses without data movement. Fabric is the strategic data platform for Azure AI Foundry in 2026.

Azure AI Search Configuration

Azure AI Search is the retrieval backbone for RAG-based agents. Key configuration decisions: index schema, chunking strategy, semantic ranking enable/disable, hybrid retrieval (keyword + vector), and re-ranking. These decisions significantly affect retrieval quality and must be made before build.

Embedding Model Selection

Azure OpenAI's text-embedding-3-large (3072 dimensions) is the recommended embedding model for production AI Foundry deployments. text-embedding-3-small (1536 dimensions) for cost-sensitive deployments. Embedding model selection affects both retrieval quality and cost.

Data Freshness

Azure AI Search supports scheduled indexer runs and push-API updates. Define your freshness requirements before configuring the indexer schedule — agents that answer questions about stale data cause user trust issues that are hard to recover from.

Access Control

Azure AI Search supports document-level security trimming using Entra ID group membership. Agents must not surface documents the querying user is not authorised to see. Security trimming must be configured deliberately — it is not on by default.

04 / Pillar 3

Use Case Prioritisation

The most expensive mistake in enterprise AI is starting with the wrong use case. Use the scoring matrix below to evaluate every candidate use case before committing to a build.

Dimension	High (3)	Medium (2)	Low (1)
Data readiness	Data in Azure, clean, indexed	Data in Azure, needs cleanup	Data outside Azure or poor quality
Query volume	>2,000 queries/month	500–2,000/month	<500/month
Decision clarity	Clear lookup / rule-based answers	Semi-structured decisions	Open-ended judgement
Labour saved	>\$100k/year in manual work	\$25k–\$100k/year	<\$25k/year
Risk level	Low-risk, reversible actions	Medium risk with human review	High risk or irreversible
M365 alignment	Copilot Studio / Teams integration	Standalone Azure agent acceptable	No Microsoft ecosystem connection

Score: 14+ = strong pilot candidate. Under 10 = address gaps first.

Azure AI Foundry agent use case patterns

Copilot Studio + AI Foundry (M365-integrated)

For organisations with M365, Copilot Studio agents that call Azure AI Foundry backends are the fastest path to user adoption — agents surface in Teams, Outlook, and SharePoint without user behaviour change.

Azure AI Foundry + Semantic Kernel (custom)

For organisations with complex orchestration needs (multi-step, multi-system, agentic loops), Semantic Kernel gives full Python/C# orchestration control with Azure AI Foundry model access and tooling.

Prompt Flow (LLMOps-first)

For organisations where operational consistency and evaluation-driven development matter as much as agent capability, Prompt Flow provides a managed DAG-based orchestration with built-in evaluation and MLOps integration.

05 / Pillar 4

Governance & Responsible AI

Microsoft's Responsible AI standard applies to all Azure AI Foundry deployments. Azure provides the most comprehensive enterprise AI governance toolkit of any cloud platform — but it must be configured. Default settings are not sufficient for production enterprise deployments.

Azure AI Content Safety

Azure AI Content Safety provides content moderation for AI agent inputs and outputs — detecting hate speech, violence, sexual content, and self-harm. Configure content safety thresholds for your use case: customer-facing agents need stricter thresholds than internal tools.

Azure Policy for AI Foundry

Azure Policy can enforce governance at the Azure AI Foundry resource level: require Private Endpoints, enforce encryption at rest with CMK (Customer-Managed Keys), restrict model deployments to approved models, and audit content safety configuration compliance.

Entra ID RBAC

Azure AI Foundry roles: AI Administrator (Hub management), AI Developer (Project creation and model deployment), AI Inference Deployment Operator (deployment management). Principle of least privilege applies — developers do not need AI Administrator access.

Customer-Managed Keys (CMK)

Azure OpenAI and Azure AI Search support CMK for data at rest via Azure Key Vault. CMK is required for HIPAA, FedRAMP High, and many financial services compliance frameworks. Configure before any production data is ingested.

Diagnostic Logging

Enable diagnostic logs for Azure OpenAI (prompt/completion logging), Azure AI Search (query logging), and Azure AI Foundry Hub (project activity). Route logs to Log Analytics for SIEM integration. Audit log retention must meet your compliance framework requirements.

06 / Pillar 5

Team Capability

Azure AI Foundry requires a specific combination of Azure platform, AI engineering, and MLOps skills. Here is an honest assessment of what is needed and how to close gaps.

Skill	What It Covers	Required For
Azure Platform Engineering	Entra ID, RBAC, Azure Networking (VNet, Private Endpoints), Key Vault, Azure Monitor	Required — all deployments
Azure AI Foundry SDK	Python azure-ai-projects SDK, Azure OpenAI SDK, Prompt Flow SDK	Required — all deployments
Azure AI Search	Index schema design, indexer config, hybrid retrieval, semantic ranking, security	Required for RAG agents
Semantic Kernel	SK agent framework, plugin design, planner configuration, memory integration	Required for complex agents
Prompt Engineering	System prompt design, few-shot examples, grounding instructions, safety instructions	Required — all agents
Prompt Flow / MLOps	DAG flow design, evaluation runs, CI/CD integration, model monitoring	Required — sustainable ops
Copilot Studio	Bot Framework, Power Platform connectors, Teams channel integration	Required — M365-integrated agents

07 / Scorecard

Self-Assessment Scorecard

For each statement tick: ✓ Fully in place / ~ Partial / ✗ Not yet

Statement	✓	~	✗
Azure Environment			
Separate resource groups for dev, staging, and production AI workloads.	■	■	■
Azure AI Foundry Hub and Project structure is defined and provisioned.	■	■	■
Managed Identity configured for AI Foundry workloads (no secrets in code).	■	■	■
Model deployments in the target region with adequate TPM quota confirmed.	■	■	■
Private Endpoints configured for Azure OpenAI and Azure AI Search (production).	■	■	■
Data Estate			
Primary data sources are in Azure (Blob, SQL, Cosmos, SharePoint, Fabric).	■	■	■
Azure AI Search index schema is designed and tested with sample queries.	■	■	■
Hybrid retrieval (keyword + vector) configured and evaluated against baseline.	■	■	■
Document-level security trimming configured for sensitive content.	■	■	■
Data freshness requirements defined and indexer schedule configured.	■	■	■
Use Case			
Candidate use cases scored using the framework in Chapter 4.	■	■	■
Pilot use case scores 14+ and has a measurable success metric.	■	■	■
Executive sponsor identified and briefed on the AI programme.	■	■	■
M365 integration path evaluated (Copilot Studio vs standalone Azure agent).	■	■	■
Governance			
Azure AI Content Safety thresholds configured for all agent surfaces.	■	■	■
CMK configured for Azure OpenAI and Azure AI Search (if required by compliance).	■	■	■
Diagnostic logging enabled and routed to Log Analytics / SIEM.	■	■	■
Azure Policy for AI Foundry applied to the Hub resource group.	■	■	■
Responsible AI impact assessment completed for the pilot use case.	■	■	■
Team			
At least one team member holds Azure AI Engineer (AI-102) certification.	■	■	■
Python Azure AI SDK and Azure AI Search SDK experience on the team.	■	■	■
Semantic Kernel or Prompt Flow experience available (in-house or partner).	■	■	■
MLOps plan in place for agent updates post-launch.	■	■	■

Score: 16–22 = Ready to build. 10–15 = Address gaps first. Under 10 = Engage a partner.

08 / Next Steps

Working with Kovil AI

Kovil AI builds production Azure AI Foundry agents for enterprise organisations — fixed-price, outcome-based.

Azure AI Foundry Strategy & Readiness

\$10,000 fixed

Azure environment audit, use case scoring, architecture blueprint. 2-week engagement.

Pilot Agent Build

\$18,000–\$32,000

One production Azure AI Foundry agent, fully deployed. Includes Azure AI Search RAG, Semantic Kernel orchestration, and post-launch support. 2–3 weeks.

Enterprise AI Foundry Programme

\$55,000–\$150,000

Multiple agents, Prompt Flow MLOps, Copilot Studio integration, enterprise governance. 6–10 weeks.

Book a readiness call: 45 minutes with a Kovil AI Azure engineer. We review your scorecard, identify the top gaps, and recommend a path to your first production Azure AI Foundry agent. kovil.ai · info@kovil.ai